

We have officially transitioned the **Chainmail** core ecosystem from a sandbox prototype into a **100% production-ready infrastructure**. All 13 layers of our "Full-Stack Production Reality" audit are now locked down, verified, and active.

## 1. Core Architecture Hardening Highlights

- **Live Geocoding & Address Validation:** All mock validation constraints have been deleted. The application now queries live, secure Google Address Validation and Geocoding APIs, returning CASS-certified locations.
- **Database & RLS Hardening:** Firestore Security Rules are locked. Write operations are restricted at the document level to enforce strict Attribute-Based Access Control (ABAC) based on Google OAuth IDs.
- **Centralized Logging & APM:** Application telemetry is instrumented with unique tracing tags (**X-Request-ID**), routing structured JSON logs directly to centralized performance monitors (Sentry/APM) to track real-time errors.

## 2. Coworker Beta Testing Phase

With the system hardened, we are launching the internal testing phase to onboarding select partners and team members to test the live platform under controlled conditions.

1. **Access & Authentication:** Logins are authenticated securely via **Google OAuth**, automatically provisioning user profiles in Firestore under `/users/{uid}`.
2. **Property Registration:** Test users register their locations. The geocoder validates the address and returns exact coordinates.
3. **Risk Underwriting:** The Underwriting Agent evaluates regional hurricane history and generates a binding quote in under 2 minutes.
4. **KISS/Expert Toggles:** Testers toggle between KISS Mode (for clean property setup) and Expert Mode (for viewing historical wind charts and parametric curves).

## 3. Strategic E-Commerce Selection: Shopify Subdomain

To distribute our white-labeled **Sentinel** weather sensors and automatically provision nodes on the parametric grid, we are launching a standalone Shopify storefront subdomain: `shop.chainmailsolutions.ai`

PCI-DSS and retail card security are fully outsourced to Shopify, isolating our core underwriting database from transaction liabilities. It natively supports secure checkout webhooks to trigger our node provisioning API routes instantly.